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1. Preamble

1.1 Purpose of the document

The purpose of this guide is to facilitate understanding of the AI Regulation from a practical perspective. To this end, it provides an introductory context for the technical guidelines, as well as hypothetical examples of high-risk systems and AI used throughout the guidelines for educational purposes to aid understanding.

This guide refers to Regulation 2024/1689 of the European Parliament and of the Council of June 13, 2024 (European Artificial Intelligence Regulation).

1.2 How to read this guide?

This guide should be read in conjunction with the Introduction to AI Regulation Guide, in order to introduce the technical guides and help clarify some of the most common questions that have arisen during participation in the Spanish Sandbox AI pilot.

The **first section** contains the preamble, which presents the general purpose of the guide.

The **second section** provides summaries of the use cases that are used as examples throughout the technical guides to illustrate their practical application.

The **third section** contains a glossary of some of the most relevant general terms and concepts (taken from Article 3 of the IA Regulation) to facilitate practical understanding of the Regulation, accompanied by examples linked to the use cases mentioned.

The **fourth section** provides context on the development of the Spanish Sandbox AI pilot, which gives rise to the technical guidelines, as well as a relationship between the guidelines and the obligations of high-risk systems, in order to facilitate consultation.

1.3 Who is it aimed at?

Companies and individuals who wish to improve their understanding of the AI Regulation and plan to use or consult the technical guidelines.

2. Examples of AI systems

This section presents a series of examples of AI systems, as patterns, which have been selected for their representativeness. Throughout all the guides and documentation provided during the sandbox, these use cases will be used as cross-cutting examples to facilitate the explanation and understanding of the different sections. All use cases presented in this section are high risk.

2.1 Biometric identification AI system at work

Title	Work attendance
Description	Artificial Intelligence biometric recognition system for recording time and attendance at work. It has cameras (or biometric sensors) that record entry, and cameras (or biometric sensors) that record exit, **without the active participation of the employee**. The system associated with each camera records entry/exit times. The intended use is to monitor time worked and not just biometric identification, so in this case, identification is not the sole purpose of the system
Responsible for potential deployment	Any company, to control and monitor the attendance of its employees.
RIA Annex	Annex III
Section	1 - Biometric systems.
Subsection	a - Biometric systems.

2.2 AI system in personnel management – Promotion

Title	Personnel Management – Promotion
Description	This artificial intelligence system evaluates the potential promotion of employees. The system collects a series of parameters to evaluate an organization's employees and determine whether or not they should be promoted to a new position. The promotion proposal provided by the AI system is one of the determining factors, rather than merely incidental, in establishing the remuneration for the new position.
Potential deployment managers	Companies that manage employee promotions, if the service or personnel management is outsourced, may have nested deployment managers.
RIA Annex	Annex III

Section	4 - Employment, worker management, and self-employment.
Subsection	b - AI intended to be used to make decisions regarding the promotion and termination of employment contracts, the assignment of tasks, and the monitoring and evaluation of the performance and conduct of individuals within the framework of such relationships.

2.3 AI system for predicting the risk of social exclusion and assessing access to aid/subsidies

Title	Prediction of social exclusion risk and assessment of access to aid
Description	This artificial intelligence system will collect specific data on the family unit, as well as general data on the assistance requested. The artificial intelligence system determines access to assistance based on specific parameters, as well as general parameters that predict the risk of short-term social exclusion. In addition to access, the system establishes a range for the amount of aid allocated. Access to aid is determined directly by the system's indication, and the amount is allocated within the range established by the system.
Potential deployment managers	Public administrations to establish access and set the amount of aid.
RIA Annex	Annex III
Section	5 - Access to and enjoyment of essential public and private services and their benefits.
Subsection	a - AI systems intended to be used by or on behalf of public authorities to assess the eligibility of natural persons for access to public assistance benefits and services, as well as to grant, reduce, withdraw, or recover such benefits and services.

2.4 AI system for chronic disease management - Smart insulin pump

Title	Chronic disease management - Smart insulin pump
Description	Artificial intelligence system that administers insulin by monitoring the patient's condition. It monitors the patient's condition by recording parameters such as blood sugar level, activity, pulse, and blood oxygen volume. These values are used by the model to predict a trend and automatically deliver insulin. The system also manages the sending of alarms to the patient and the doctor.
Potential deployment managers	The medical institution or hospital that provides the system to the physician, who will then apply it to patients, would be responsible for deploying the system.
RIA Annex	Annex I
Section	A
Subsection	11 - Medical devices.

2.5 AI system for detecting false reports

Title	Detection of false claims
Description	Artificial intelligence system used to determine the probability that a complaint filed at a police station is false, based on the complaint filed by the citizen and transcribed into the system by them or the police.
Potential deployers	Security forces (both national and regional in each case) that manage and process reports from citizens.
RIA Annex	Annex III
Section	6 - Law enforcement matters.
Subsection	c - AI systems intended for use by law enforcement authorities for assessing the reliability of evidence during the investigation or prosecution of criminal offenses.

3. Definitions with examples

This section addresses some of the general concepts and terms described in Article 3 of the most relevant Regulation. All of these terms and concepts are used throughout the rest of the sandbox guides and materials.

This content is intended to serve as an initial introduction to the terms and as a reference throughout the sandbox implementation. Most definitions are accompanied by examples applicable to the AI systems described above, with the aim of providing context for the terms presented.

The definitions provided refer to the version of the Regulation published by the Council of the European Union on June 13, 2024. Only one of these definitions is not included in the Regulation, namely the concept of the "life cycle of an AI system," which complements the overall understanding of the stages of a system.

3.1 Artificial intelligence system (AI system)

Artificial intelligence system: a machine-based system that is designed to operate with varying levels of autonomy and may exhibit adaptive capabilities after deployment, and which, for explicit or implicit purposes, infers from input data the generation of output results, such as predictions, content, recommendations, or decisions, that may influence physical or virtual environments.

The European Commission has published guidelines aimed at clarifying this definition of an AI system[(1).]

Example - Biometric identification at work
A set consisting of cameras that capture information and a model that analyzes the data and compares it with stored information to generate identification, record information about entry or exit events, and thus monitor time worked. This example would be a high-risk AI system as regulated in Annex III, specifically in section 1.

Example - Chronic disease management - Smart insulin pump
Consisting of a mechanism for collecting data on blood glucose concentration which, together with other attributes stored in the patient database (patient physiology, history, and activities such as meals and physical activity), analyzes and determines the correct dose of insulin to be administered. This example would represent a high-risk AI system as regulated in Annex I, specifically falling under the sectoral legislation on medical devices mentioned in section 11 of that Annex.

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https://ec.europa.eu/newsroom/dae/redirection/document/118640

3.2 Placing on the market

Placing on the market: the first making available on the Union market of an AI system or an AI model for general use;

Example
In the defined use cases, the placing on the Union market of the AI systems described would correspond to the first time that such a system is put into service and used, whether developed or imported by a natural or legal person physically present or established in the Union.

3.3 Placing on the market

Marketing: the supply of an AI system or an AI model for general use for distribution or use on the Union market in the course of a commercial activity, whether in return for payment or free of charge.

In the defined use cases, the marketing of the AI systems described would correspond to the moment when such systems are distributed or used on the market within the Union, whether for remuneration or free of charge. This means that any of the potential parties responsible for deployment could acquire and implement it.

Example – Chronic disease management – Smart insulin pump
When it is available and the supplier has been contracted by those responsible for deployment, such as hospitals, to distribute it to their patients.

3.4 Intended purpose

Intended purpose: the use for which a supplier intends an AI system to be used, including the specific context and conditions of use, as provided by the supplier in the instructions for use, promotional and sales materials, and technical documentation.

Example – Biometric identification at work
Its intended purpose would be to identify employees in order to record their arrival and departure times, by extracting biometric patterns through the entry/exit cameras and comparing them with previously registered identities. Additionally, it should be explained that this system performs a complete scan of the employee's face and that, once this data has been identified and collected, the system will compare it with the information stored in the database and analyze whether the features it has identified correspond to any of those stored in the database, identifying the person responsible for the deployment and recording the corresponding entry or exit event.

3.5 Security component

Safety component: a component of an AI product or system that performs a safety function for that AI product or system, or whose failure or malfunction endangers the health and safety of persons or property;

Example – Chronic disease management – Smart insulin pump

The AI system collects data on blood glucose concentration, which it analyzes together with other attributes stored in the patient's database, and determines the correct dose of insulin to be administered. In this context, the safety of these systems is crucial, as any failure could endanger the patient's life. An example of a safety component is a sensor that warns of insulin pump failure in the event that it is not operating properly or is even inoperative. The system will be equipped with a large number of safety components, but all of them must clearly notify the person responsible for deploying the device of any malfunction.

3.6 Substantial modification

Substantial modification: a change to an AI system after its introduction on the market or into service that was not anticipated or planned in the initial conformity assessment carried out by the supplier and as a result of which the AI system's compliance with the requirements set out in Chapter III, Section 2 is affected, or which results in a change to the intended purpose for which the AI system in question has been assessed;

In any of the use cases presented, a substantial modification will be considered to be a change in the purpose or intent of the system to another high-risk purpose. Similarly, any modification to the system that could affect compliance with the requirements of the Regulation that have previously been assessed for conformity (risk management system, data and data governance, technical documentation, event logging, transparency vis-à-vis those responsible for deployment, human oversight, accuracy, robustness, and cybersecurity) will be considered substantial.

Example – Detection of false reports

The initial purpose of the system is to determine the probability that a complaint filed at a police station is false. A substantial modification could occur if the system were used in investigations to assess the veracity of statements made by individuals under investigation. Or if texts from social media interactions were entered into the system to determine patterns of falsehood in these contexts.

3.7 Post-marketing surveillance system

Post-market surveillance system: all activities carried out by AI system providers to collect and review experience gained from the use of AI systems they place on the market or put into service, with the aim of detect the possible need to immediately apply any type of corrective or preventive measure that may be necessary.

Example – Chronic disease management – Smart insulin pump

The system provider shall have a procedure in place for post-market monitoring of the product, collecting data and information on the performance of the system and feedback from those responsible for its deployment, enabling it to identify, for example, possible errors in its operation and correct them. This post-marketing monitoring will involve, where appropriate, the notification of incidents to the market surveillance authority and the assessment of possible substantial modifications to the artificial intelligence system.

3.8 Common specification

Common specification: a set of technical specifications as defined in Article 2, point 4, of Regulation (EU) No. 1025/2012 that provides a means of meeting certain requirements established under this Regulation;

In general, and in accordance with Regulation (EU) No. 1025/2012, a technical specification is a document that prescribes the technical requirements that a product, process, service, or system must meet. The technical specification refers to aspects such as quality, performance, interoperability, environmental protection, health and safety, product dimensions, naming, testing, testing methods, packaging, marking or labeling, and conformity assessment procedures.

In the context of the Regulation, and with regard to common specifications, it is stated that:

1. If **harmonized standards** do not exist (or when the Commission considers them insufficient), the Commission may adopt **common specifications** in relation to the **requirements of Chapter 3, Section II**.
2. When developing the **common specifications**, the Commission **shall seek** the **views** of relevant expert bodies or groups (established in accordance with applicable Union law at sectoral level).
3. High-risk AI systems that comply with the **common specifications** shall be presumed to comply with the **requirements of Chapter 3, Section II** (to the extent that those common specifications cover those requirements).
4. Where providers do not comply with the **common specifications**, they shall duly justify that they have adopted **technical solutions** that are at least **equivalent** to those specifications.

Common specifications are drawn up by the European Commission. For example, there are common specifications in Regulation 2017/745 of April 5 on medical devices, to implement the requirements for certain medical devices.

3.9 Training data

Training data: data used to train an AI system by adjusting its trainable parameters.

Example – Prediction of risk of social exclusion and assessment of access to aid/subsidies

The data used to train the system is the stored data associated with different quantitative parameters (income level, education level, occupation, geographical areas, etc.) and qualitative parameters (standardized reports, textual reports). In other words, all the data on benefits—whether granted or not—that are selected to develop the artificial intelligence system, as data used during its training.

3.10 Validation data

Validation data: data used to provide an assessment of the trained AI system and adapt its non-trainable parameters and learning process to, among other things, avoid underfitting or overfitting;

Example – Prediction from risk from exclusion social and assessment from access to aid/subsidies

Validation data is data that allows us to evaluate the system we have trained. This data also belongs to the stored data associated with different quantitative parameters (income level, education level, occupation, geographical areas, etc.) and qualitative parameters (standardized reports, textual reports). It can be a subset of the data used in training or another subset of data from that history that was not used in the training phase.

3.11 Test data

Test data: data used to provide an independent evaluation of the AI system, in order to confirm the intended functioning of that system before it is placed on the market or put into service.

Example – Prediction from risk from exclusion social and assessment from access to aid/subsidies

Test data is data that allows us to independently evaluate the trained and validated system. This dataset must be different from the one used for training and validation, but must be of exactly the same nature: stored data associated with different quantitative parameters (income level, education level, occupation, geographical areas, etc.) and qualitative parameters (standardized reports, textual reports).

3.12 Input data

Input data: data provided to an AI system or obtained directly by it, from which an output result is produced;

Example – Prediction from risk from exclusion social and assessment from access to aid/subsidies

Input data is data that is entered into the system once it has been trained, validated, tested, and put into service. In this case, it would be all the information collected for the application process by the person responsible for deployment, as well as reports that may be provided or requested from other entities (social protection, primary care, etc.). Similarly, as this is a public administration system, it could use input data from other registered sources, such as data from the tax authorities, land registry, etc.

3.13 Biometric identification

Biometric identification: the automated recognition of physical, physiological, behavioral, or psychological characteristics to determine the identity of a natural person by comparing their biometric data with the biometric data of persons stored in a database.

Example – Biometric identification at work

The act of recognizing employees' biometric characteristics through cameras located at the entrance to the premises. In this context, this AI system is a biometric identification system, which may or may not be considered high risk depending on its use.

3.14 Biometric verification

Biometric verification: automated, one-to-one verification, including authentication, of the identity of individuals by comparing their biometric data with previously provided biometric data.

Example – Biometric identification only to identify the identity of a worker when accessing the workplace.

The automated comparison of identified biometric data against reference data in the company's database. In this context, this AI system is a biometric identification system but not a high-risk one when used solely to allow access to the company.

3.15 Biometric categorization system

Biometric categorization system: an AI system intended to include individuals in specific categories based on their biometric data, unless it is ancillary to another commercial service and strictly necessary for objective technical reasons.

Example

An AI system categorizes website users and potential customers based on their behavior on the website, through an analysis of their keystrokes.

3.16 Remote biometric identification system

Remote biometric identification system: an AI system designed to identify individuals without their active participation and generally at a distance by comparing their biometric data with that contained in a reference database.

In other words, the AI system aims to detect multiple individuals or their behavior simultaneously, thereby significantly simplifying the identification of individuals without their active participation. This excludes AI systems intended for biometric verification, including authentication, whose sole purpose is to confirm that a specific natural person is who they claim to be, as well as the identity of a natural person for the sole purpose of granting them access to a service, unlocking a device, or granting them security access to premises.

3.17 Serious incident

Serious incident: an incident or malfunction of an AI system that, directly or indirectly, has any of the following consequences:

a) the death of a person or serious damage to their health;

b) a serious and irreversible disruption to the management or operation of critical infrastructure;

c) breach of obligations under Union law aimed at protecting fundamental rights;

d) serious damage to property or the environment;

Example – Chronic disease management – Smart insulin pump
The AI system collects data on blood glucose concentration and, together with other attributes stored in the patient database (patient physiology, history, activities such as meals and physical activity), analyzes and determines the correct dose of insulin to be administered. Expanding on the example of the definition of withdrawal from the artificial intelligence system. In this scenario, a system failure could result in excessive or insufficient insulin administration. If such a failure resulted in death or serious damage to health, the incident would clearly be considered serious.

3.18 Real-world testing plan

Real-world testing plan: a document describing the objectives, methodology, geographical, population, and time scope, monitoring, organization, and implementation of real-world testing.

3.19 Deepfake

Deepfake: image, audio, or video content generated or manipulated by AI that resembles real people, objects, places, entities, or events and may mislead a person into believing that it is authentic or true.

3.20 General-purpose AI model (GPAI)

General-purpose AI model: an AI model trained on a large volume of data using large-scale self-supervision, which has a considerable degree of generality and is capable of competently performing a wide variety of different tasks

, regardless of how the model is introduced to the market, and which can be integrated into various downstream systems or applications, except for AI models used for research, development, or prototyping activities prior to their introduction to the market.

Requirement to be considered a GPAI: its computational training cost must be greater than 10^{23}. $FLOPs$

Example

From OpenAI models such as: GPT-4, GPT-3; from Meta, the LLaMA family of models; from Google, Gemini Ultra or Gemini Pro; from Mistral, Mistral Large 2 or Mistral Small; from Anthropic, Claude 3 Opus or Claude 3 Haiku; among others. It must be distinguished from general-purpose AI systems such as ChatGPT or Copilot.

3.21 Systemic risk of general-purpose AI models

Systemic risk: Some general-purpose AI models may pose an additional risk due to their high-impact capabilities. They may have a significant impact on the Union market due to their scope or actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or society as a whole, which may spread on a large scale throughout the value chain.

Requirement to be considered a GPAI with systemic risk: its computational training cost must be greater than > $10^{(25)}$. $FLOPs$

3.22 Life cycle of an AI system

Life cycle of an AI system: a structured and well-defined sequence of stages that software goes through during its lifetime, from the definition of initial requirements (design inputs) to its withdrawal, including the stages of design and development, validation and verification testing, deployment, operation and monitoring, reassessment, and continuous validation.

4. Introduction to Technical Guidance

This section provides context on the development and purpose of the technical guidelines, which were drawn up as part of the Spanish AI sandbox pilot project promoted by the Secretary of State for Digitalization and Artificial Intelligence with the support of the European Commission.

The objective of the sandbox and all the materials developed and made publicly available is to help the Spanish business community advance in AI regulatory compliance, providing support to companies for the commercialization of high-quality AI systems and increasing public confidence in the use of this technology.

4.1 The Spanish AI Sandbox pilot

The Spanish AI Sandbox pilot is a controlled testing environment that **seeks to promote responsible AI development by validating the compliance of high-risk systems with the requirements of the Regulation through conformity assessment simulations**. Its main objective is to clarify the obligations of the Regulation for startups and SMEs, validate the usefulness of technical guidelines, and facilitate knowledge transfer to create solutions aligned with legal and ethical principles. In addition, it drives reliable innovation, strengthens the supervisory capacity of the Spanish AI Supervisory Agency, and allows for the assessment of the real impact of regulatory obligations, contributing to European standardization. Ultimately, it aims to position Spain as a leader in responsible AI at the European and global levels.

In November 2023, the Secretary of State for Digitalization and AI, in close collaboration with the European Commission (DG CNECT), published Royal Decree 817/2023, which establishes the regulatory framework for companies to participate in the AI sandbox. Subsequently, in December 2024, the official call for proposals was launched by resolution. In January 2025, the call for proposals was presented at an event.

After evaluating the 41 proposals submitted, 12 high-risk systems in different sectors were selected:

Illustration 1: Sandbox participants

Participantes

Sector	Descripción	Sistema	Colaborador	Usuario
Acceso a servicios y ayudas	Sistema de procesado de **la transcripción inteligente de las comunicaciones de emergencias** del centro de operaciones del 112 de Galicia	citic	CLUSTER TIC GALICIA, cinfo	axega
	Sistema de IA para la **calificación crediticia, análisis de solvencia y riesgo de impago** de individuos a partir de sus datos transaccionales bancarios	DEDOMENA		BBVA, ikualo, ZIEN
Biometría	**BioSurveillance** - Solución de **videovigilancia de alto rendimiento**, que funciona por reconocimiento facial (mediante biometría). Está especialmente diseñada para la **identificación simultánea de sujetos en entornos multitudinarios y muy cambiantes**	herta		
	Veridas Age Verification - Sistema para **verificar la edad de usuarios en máquinas de vending**, asegurando el cumplimiento normativo en la venta de productos regulados como tabaco o también normativa pendiente de regular para vapeadores	VeriDas	GM Global Solutions	UAM
Empleo	**Shakers AI Matchmaker** - Sistema de emparejamiento impulsado por IA que utiliza recomendaciones personalizadas para **conectar talento freelance con proyectos empresariales**	SHAKERS		BBVA
	Ranking de inscritos vía HyM (How you Match) - Adecuación a la vacante mediante el **cálculo de compatibilidad entre los currículums de los candidatos** y los requisitos de las ofertas de empleo en InfoJobs	Adevinta	Hirint.	SIETE ESTRELLAS, Proselección, on SOLUCIONES

Prior to the start of the project, technical guidelines had been developed, precursors to these guidelines, with the aim of facilitating understanding of the European AI Regulation and the obligations applicable to high-risk systems for participants in the sandbox. These guidelines are not binding and do not replace the regulations, but offer practical recommendations in line with regulatory requirements.

In April 2025, a **kick-off** event was held to present the working method to the selected participants, along with the proposed work schedule and the main milestones to be achieved. In summary, participation is structured in five phases:

Illustration 2: Sandbox Phases

To support the development of sandbox activities, on June 10, 2025, a **group of 40 expert advisors** was appointed by resolution. This group is made up of independent professionals of recognized prestige and technical experience in related fields of knowledge, with present or past responsibilities in academia and universities, professional associations, other types of associations, or the business sector. Its objective is to contribute technical knowledge, support participants based on their experience, and prepare training material that can be used by Spanish companies in their regulatory compliance initiatives.

4.2 Context of publication of the technical guides

The technical guidelines drawn up within the framework of the Spanish AI Sandbox pilot have been developed thanks to collaboration between the Secretary of State for Digitalization and Artificial Intelligence, as the driving force behind the initiative, and various stakeholders: participants, technical assistants, the group of expert advisors, and various potential national authorities responsible for the Regulation.

During their preparation and use, it has been noted that, during the process of developing common standards and specifications for the AI Regulation, **they are helping participating companies to advance their initiatives aimed at ensuring future regulatory compliance, as well as improving their internal quality and safety systems** in the development of AI.

That is why it has been estimated that the publication of this material could be extremely useful for the Spanish business community as a whole, especially for SMEs and start-ups. As indicated on the back cover of each guide, **the guides are not binding and do not replace or develop the applicable regulations**; **they** simply **provide practical recommendations in line with regulatory requirements pending the approval of harmonized rules applicable to all member states**.

Below is a summary of the obligations and functions established by the AI Regulation for suppliers and deployment managers covered in the technical guides:

Operator	Obligation	AI Regulation	Associated guide
HRIA provider	Compliance with requirements section 2 of the Regulation	Art. 16.a	15. Technical Documentation Guide 11. Cybersecurity Guide 7. Data and Data Governance Guide 5. Risk Management Guide 9. Accuracy Guide 10. Soundness Guide 8. Transparency Guide

			6. Human Oversight Guide
HRIA Provider	Count with a quality management system quality management system	Art 16.c	4. Quality Management Guide
HRIA Supplier	Retain files of records automatically generated by your systems	Art. 16.e	12. Records Guide
HRIA Provider	Evaluation procedure Co mpliance	Art 16.f	3. Compliance Assessment Guide
HRIA Provider	Adopt necessary corrective measures	Art 16.j	14. Incident reporting guide
HRAI Providers	Incident of incidents	Art 73	14. Incident reporting guide

Operator	Obligation	IA Regulations	Associated guide
Responsible for deployment HRIA	Assign the human oversight of the system to individuals with competence, training and authority necessary.	Art. 26.2	6. Human supervision guide
Responsible for deployment HRIA	Retain files of automaticall y generated records by your systems	Art 26.6	12. Record keeping guide
Responsible for deployment HRIA	Report the supplier or distributor if they believe that use of the system in accordance with their instructions may pose a risk to health, safety, or fundamental rights , and	Art 26.5	14. Incident reporting guide

	suspend use of the system if applicable		
Responsible for deployment HRIA	Inform the supplier or distributor, importer, and authority of market surveillance serious incidents.	Art 26.5	14. Guidance on reporting incidents